# Linhao Bai

Robotics Systems | Robot Learning | Distributed Robotics | AI

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## EXPERIENCE

### Tesla | Robotics Integrations Engineer
Sep 2026 - Present

Austin, TX | Mobile Robotics / Overmind

- Work on production AMR, AGV, and ASRS systems, spanning multi-robot fleet orchestration, job/task sequencing, traffic coordination, and robot-to-factory workflows.
- Analyze system behavior across distributed services, Kubernetes, NATS messaging, telemetry, robot runtime, localization, and PLC / OPC UA interfaces, connecting software state to physical operations.
- Work with simulation and virtual commissioning to validate robotic tasks, workflow transitions, industrial equipment interfaces, and factory automation scenarios.
- Design AI-agent-assisted diagnostic workflows connecting incident evidence, logs, system configuration, and code context to make robotics troubleshooting more repeatable.

### Georgia Tech | Graduate Research Assistant, STAR Lab
Oct 2024 - May 2026

Atlanta, GA

- Researched dexterous robot learning, visuomotor dynamics, policy learning, and sim-to-real transfer; co-authored K-UBM (CoRL 2026) and Video2Sim2Real.
- Built ROS-based real-time learned-policy execution on a Kinova arm and dexterous hands, including joint command streaming, synchronized control, and state feedback.

### SEER Robotics | Software Engineer / Project Manager
Jun 2022 - Jul 2024

Shanghai, China

- Developed C++ robot-controller utilities for OTA / backup, time synchronization, health monitoring, telemetry, and hardware / software interfaces.
- Worked on international AMR systems and a five-robot, 100+ rack warehouse system involving fleet scheduling, reservations, queueing, SLAM / QR correction, and randomized robot tests.

## RESEARCH & PUBLICATIONS

**K-UBM | CoRL 2026 | Equal contribution**
Going with the Flow: Koopman Behavioral Models as Pseudo Planners for Visuo-Motor Dexterity.

**Video2Sim2Real | Preprint 2026**
Full-Stack Autonomous Dexterous Skill Acquisition from a Single Human Video.

**IEEE Signal Processing Letters | 2026**
A 1/R Law for Kurtosis Contrast in Balanced Mixtures.

## SELECTED PROJECTS

**Multi-Robot Planning & Scheduling Simulation**
Reproduced multi-agent pathfinding and coordination methods in simulation, examining path conflicts, robot interactions, and scheduling behavior.

**SO-101 & MicroDuck | Robot Learning Reproductions**
Reproduced open-source manipulation and locomotion-learning methods, exploring imitation learning, reinforcement learning, policy training, and evaluation.

## EDUCATION

Georgia Institute of Technology | M.S. Computer Science, GPA 4.0 / 4.0 | 2026

Chongqing Jiaotong University | B.Eng. Water Resources Engineering | 2022

## SKILLS

Python, C++, PyTorch, ROS / ROS2, MuJoCo, Linux, Docker, Kubernetes, Git, NATS, OPC UA, SLAM, multi-agent planning, imitation learning, reinforcement learning, Koopman dynamics, sim-to-real.